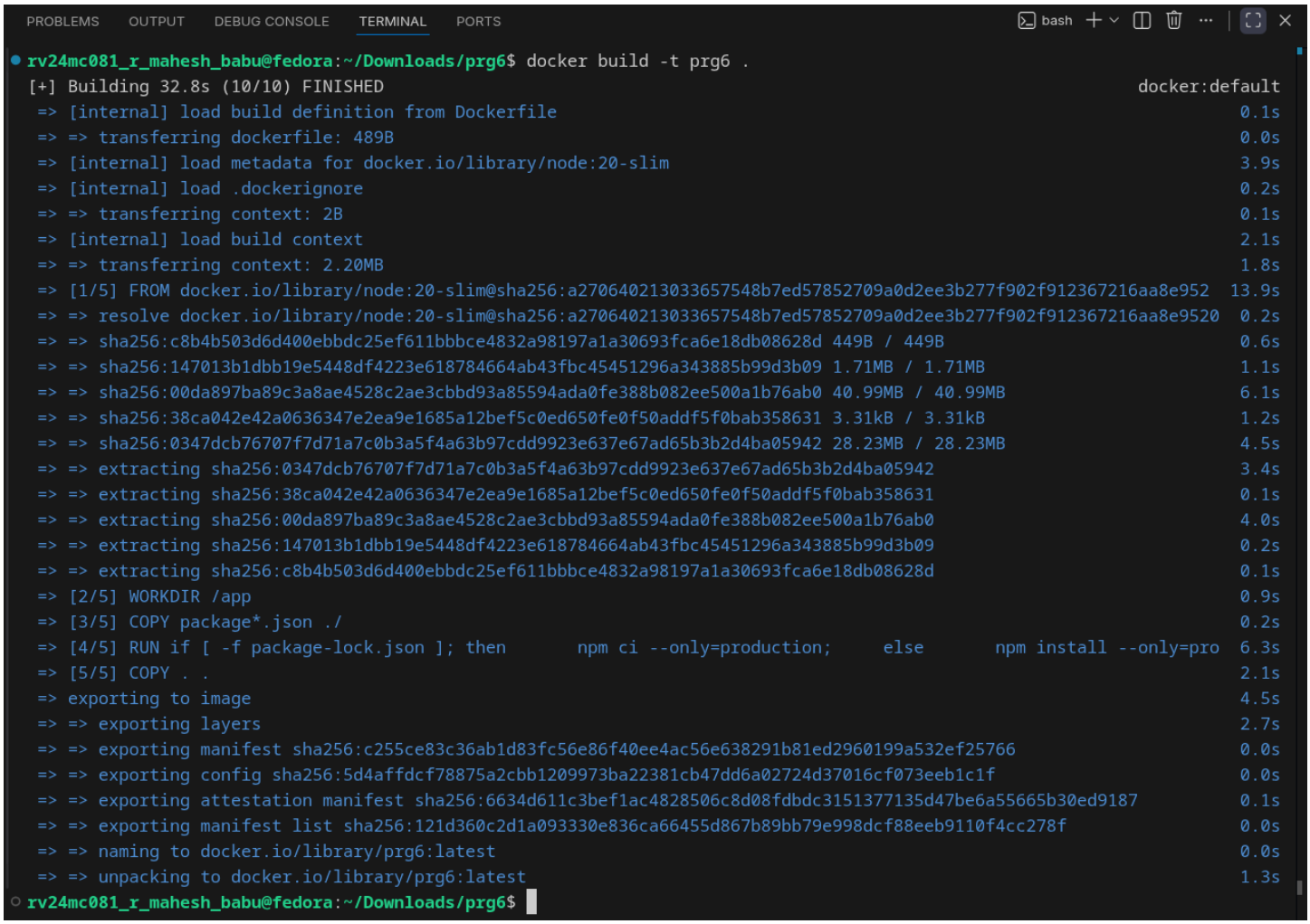
Program-6

Title: Implementing an Automated Build Pipeline Using Docker Hub

This document describes the step-by-step process of developing an application, containerizing it with Docker, and publishing the resulting image to Docker Hub and the GitHub Container Registry (GHCR). This workflow establishes a basic yet effective CI/CD pipeline.

1. Local Project Setup

The process begins with creating the project directory and preparing the local development environment. This step includes organizing files and initializing the application source code required for containerization.



2. Version Control and Source Code Management (Git & GitHub)

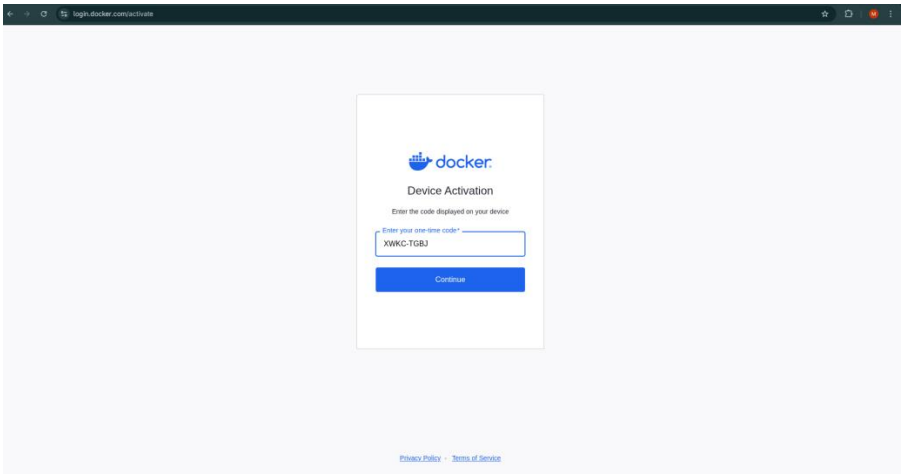
The application source code is tracked using Git and pushed to a remote GitHub repository. This enables version control, collaboration, and serves as the foundation for future automated build processes.

3. Building the Docker Image Locally

In this stage, the application along with its runtime environment—defined within the Dockerfile—is packaged into a single, executable Docker image. This image contains all dependencies and configurations needed to run the application consistently across different systems.

Login with Dockerhub credentials and with github credentials

```
rv24mc081_r_mahesh_babu@fedora:~$ docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <
username>'


Your one-time device confirmation code is: XWKC-TGBJ
Press ENTER to open your browser or submit your device code here: https://login.
docker.com/activate

Waiting for authentication in the browser…

WARNING! Your credentials are stored unencrypted in '/home/rv24mc081_r_mahesh_ba
bu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
rv24mc081_r_mahesh_babu@fedora:~$
```




```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                    bash

rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg6$ docker push mahesh158/prg6:latest
The push refers to repository [docker.io/mahesh158/prg6]
0347dcb76707: Pushed
6aa6c21511ff: Pushed
7b35c2726f0b: Pushed
c8b4b503d6d4: Pushed
8369cc62f7e5: Pushed
d8d1bb2199c7: Pushed
38ca042e42a0: Pushed
00da897ba89c: Pushed
147013b1dbb1: Pushed
1e882b85561c: Pushed
latest: digest: sha256:121d360c2d1a093330e836ca66455d867b89bb79e998dcf88eeb9110f4cc278f size: 856
rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg6$ docker tag prg-6 ghcr.io/R Mahesh Babu/prg6:latest
```

```
uuJfMW5nMc3WVpBV | ghcr.io -u /rvcestudent --password-stdin

WARNING! Your credentials are stored unencrypted in '/home
hesh Babu-p/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg6$
rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg6$ echo  ghp_2Vf4d2p3blSYjj
rv24mc081_r_mhesh_babu@fedora:~/Doownloads/prg6'/home/IRV2       R Mahesh
Babu-p/.docker/config.json'.   ~/Downloads/prg6$ docker push ghcr.io/rvces
tudent/program-6:latest
```